

Week 5 HW Question 9:

Date:

We want scalars a and b such that $au + bw = v$

When $u = (3, 2)$, $w = (3, -3)$, $v = (6, -1)$.

This gives us the system $a(3, 2) + b(3, -3) = (6, -1) \rightarrow \begin{cases} 3a + 3b = 6 \\ 2a - 3b = -1 \end{cases}$

$$\left[\begin{array}{cc|c} 3 & 3 & 6 \\ 2 & -3 & -1 \end{array} \right]$$

$$1.) \frac{1}{3} R_1 \rightarrow R_1 \quad \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 2 & -3 & -1 \end{array} \right]$$

$$2.) -2R_1 + R_2 \rightarrow R_2 \quad \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -5 & -5 \end{array} \right]$$

$$3.) -\frac{1}{5} R_2 \rightarrow R_2 \quad \left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \end{array} \right]$$

$$4.) -R_2 + R_1 \rightarrow R_1 \quad \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array} \right]$$

So, $a = 1$ and $b = 1$

Therefore, $v = 1u + 1w$
 $= u + w$